

```

✓ /*****
  * Task 1: Read array values and sort them
  *****/

  // Get array size from user
  const size = Number(prompt("Enter the size of the array"));
  const arr = new Array(size);

  // Fill array values
  ✓ for (let i = 0; i < size; i++) {
  ✓   arr[i] = Number(prompt(`Enter the value at index ${i}`));
  }

  // Display original and sorted arrays
  console.log("Original Array:", arr);
  ✓ console.log("Ascending Array:", [...arr].sort((a, b) => a - b));
  ✓ console.log("Descending Array:", [...arr].sort((a, b) => b - a));
  ,

```

```

✓ /*****
  * Task 2: Address object and formatter function
  *****/

  // Create address object
  ✓ const address = {
    street: prompt("Enter street"),
    buildingNum: prompt("Enter building number"),
    city: prompt("Enter city"),
  };

  // Function to format address with current date
  1 reference
  ✓ function showAddress(addr) {
    const today = new Date();
    ✓ const date = `${today.getDate()}/${today.getMonth() + 1}/${today.getFullYear()}`;

    ✓ return `${addr.buildingNum} ${addr.street}, ${addr.city} city registered on ${date}`;
  }

  // Display formatted address
  console.log(showAddress(address));

```

```

/*****
 * Task 3: Display object value based on key
 *****/

// Function to display value with validation
2 references
function displayValue(obj, key) {
  if (!(key in obj)) {
    return `${key} key not found in the given object`;
  }

  if (key === "age") {
    return `${obj[key]} years old`;
  }

  return obj[key];
}

// Person object
const person = {
  name: "Ahmed",
  age: 22,
};

// Test function
console.log(displayValue(person, "name"));
console.log(displayValue(person, "age"));

```

```

/*****
 * Task 4: Auto scrolling window
 *****/

let advWindow;

// Open a new window and auto-scroll vertically
0 references
function selfScroll() {
  advWindow = window.open(
    "https://www.google.com",
    "_blank",
    "width=400,height=400,top=100"
  );

  let scrollPosition = 0;
  let direction = 1; // 1 = down, -1 = up
  const step = 20;
  const delay = 30;

  2 references
  function autoScroll() {
    // Stop if window is closed
    if (!advWindow || advWindow.closed) return;

    const maxScroll =
      advWindow.document.body.scrollHeight - advWindow.innerHeight;

    scrollPosition += step * direction;
    advWindow.scrollTo(0, scrollPosition);

    // Reverse direction at limits
    if (scrollPosition >= maxScroll) direction = -1;
    if (scrollPosition <= 0) direction = 1;

    setTimeout(autoScroll, delay);
  }

  autoScroll();
}

```